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All VS are $\mathbb{R}$-VS in 247

Recall 146

Def Let X be a VS, a norm on X is a map $\|\cdot\|: X \to \mathbb{R}$ s.t.

1) $\forall a \in \mathbb{R}, \forall x \in X$, $\|ax\| = |a|\|x\|$
2) $\forall x \in X$, $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$
3) (Triangle ineq) $\|x+y\| \leq \|x\| + \|y\|$

All finite dimensional VS have a norm (usually many)

Ex. (Norms)

- $\mathbb{R}^n$: $\|x\|_1 = \sum_{i=1}^{n} |x_i|$ (ℓ^1-norm)
- $\mathbb{R}^n$: $\|x\|_\infty = \max_k |x_k|$ (ℓ^∞-norm)
- $\mathbb{R}^n$: $1 \leq p < \infty$ $\|x\|_p = \left(\sum_{k=1}^{n} |x_k|^p\right)^{\frac{1}{p}}$ (ℓ^p-norm)

Def Let $(X, \|\cdot\|)$ be a NVS, the distance function of X is $d: X \times X \to X$ with $d(x,y) = \|x-y\|$.
- $d(x,y) = d(y,x)$
- $d(x,y) \geq 0$, $d(x,y) = 0 \iff x = y$
- $d(x,y) \leq d(x,z) + d(y,z)$

Def An inner product on X is $\langle \cdot, \cdot \rangle: X \times X \to \mathbb{R}$ s.t.
- Bilinear
- Symmetric
- $\langle x, x \rangle \geq 0$ w/eq $\iff x = 0$

Ex
- $X = \mathbb{R}^n$, $\langle x, y \rangle = x^T y = \sum_{k=1}^{n} x_k y_k$

Prop Let $\langle \cdot, \cdot \rangle$ be an IP on X. Define $\|\cdot\|: X \to \mathbb{R}$ by $\|x\| = \sqrt{\langle x, x \rangle}$. Then $\|\cdot\|$ is a norm

Pf Obvious from Cauchy-Scharz. □

Thm (Cauchy-Scharz inq) Let $\langle \cdot, \cdot \rangle$ be an IP on X. Let $\|\cdot\|: X \to \mathbb{R}$, $x \mapsto \sqrt{\langle x, x \rangle}$. Then $\forall x, y \in X$,
$$|\langle x, y \rangle|^2 \leq \|x\| \cdot \|y\|$$
w/eq iff $\{x, y\}$ is linearly dependent.

Pf Let $v, w \in X$, WLOG $x, y \neq 0$. Let $t \in \mathbb{R}$

$$0 \leq \|v - tw\|^2 = \langle v - tw, v - tw \rangle$$
$$= \|v\|^2 - 2t\langle v, w \rangle + t^2\|w\|^2$$

$at^2 + bt + c \geq 0$ where
$a = \|w\|^2$ $b = -2\langle v, w \rangle$ $c = \|w\|^2$,

thus $0 \leq b^2 - 4ac \implies 4\langle v, w \rangle^2 \leq 4\|v\|^2\|w\|^2$

w/eq $\iff \|v - tw\|^2 = 0 \iff v = tw \iff \{v, w\}$ lindependent □

Ex
- $\mathbb{R}^{m \times n}$ $\langle A, B \rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij} B_{ij} = \sum_{i=1}^{m}\sum_{j=1}^{n} (A^T)_{ji} B_{ij} = \sum_{j=1}^{n} (A^T B)_{jj} = tr(A^T B) = tr(B^T A) = tr(BA^T) = tr(AB^T)$

  induces $\|A\|^2 = \langle A, A \rangle = tr(A^T A) = tr(AA^T)$ (Frobenius norm)

Prop Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $Ax \in \mathbb{R}^m$, $(Ax)_i = \sum_{j=1}^{n} A_{ij} x_j$
Then $\|Ax\| \leq \|A\| \cdot \|x\|$

Pf
$$\|Ax\|^2 = \sum_{i=1}^{m} (Ax)_i$$
$$= \sum_{i=1}^{n} \langle A_i, x \rangle^2$$
C-S $$\leq \sum_{i=1}^{n} \|A_i\|^2 \|x\|^2$$
$$= \|A\|^2 \|x\|^2$$ □

Def Let X, Y be NVS, $T \in \mathcal{L}(X, Y)$ is bounded iff $\exists c > 0$ s.t. $\|T(v)\| < c\|v\|$, $\forall v \in X$

Ex.
- All $A \in \mathbb{R}^{m \times n}$ are bounded.
- $\mathcal{B}(X, Y)$ set of all bounded linear maps $X \to Y$ (depends on norm)
  $\mathcal{B}(X, Y)$ is a subspace of $\mathcal{L}(X, Y)$ with the operator norm
  $\|T\|_{op} = \inf\{C > 0 : \|T(v)\| \leq C\|v\|, \forall v \in X\}$

Topology of normed vectors spaces

Let X be a NVS. Recall $d: X \times X \to \mathbb{R}^{\geq 0}$, $d(v, w) = \|v, w\|$.

Def Let $v \in X$, $r > 0$. The open ball with center v and radius r is
$B_r(v) = \{w \in X : d(v, w) < r\}$

Def Let $v \in X$, $r > 0$, closed ball
$\overline{B_r(v)} = \{w \in X : d(w, v) \leq r\}$

Def Let $U \subseteq X$, U is open iff
$\forall v \in U, \exists \varepsilon > 0$ s.t. $B_\varepsilon(v) \subseteq U$

Lemma i) if $U_\alpha \subseteq X$ is open then $\bigcup_{\alpha \in A} U_\alpha$ is open
ii) if $U_1, \ldots, U_n$ are open $\bigcap_{k=1}^{n} U_k$ is open

Def Let $F \subseteq X$, F is closed iff F^c is open

Prop Closed balls are closed sets

Pf $(\overline{B_r(v)})^c = X \setminus \overline{B_r(v)} = \{w \in X : d(v, w) > r\}$
if $w \in (\overline{B_r(v)})^c$, $\|v - w\| > r$
Let $\varepsilon = \|v - w\| - r > 0$
Let $u \in B_\varepsilon(w)$
$d(u, w) < \varepsilon$
$d(v, w) \leq d(v, u) + d(u, w) < d(u, v) + \varepsilon$
$d(u, v) > d(v, w) - \varepsilon = r$
Thus $u \in (\overline{B_r(v)})^c$ so $B_\varepsilon(w) \subseteq (\overline{B_r(v)})^c$
So $(\overline{B_r(v)})^c$ is open □

Lemma i) If $F_\alpha \subseteq X$ is closed $\forall \alpha \in A$, then $\bigcap_{\alpha \in A} F_\alpha$ is closed
ii) If $F_1, \ldots, F_n \subseteq X$ are closed then $\bigcup_{n=1}^{N} F_n$ is closed

Pf Opens sets and deMorgans laws □

Def Let $A \subseteq X$. The interior of A (A° or $int(A)$) is defined to be
$A^\circ = \bigcup_{V \subseteq A} V$ (V is open) (Note that this we defined)

Prop 1) A° is open and $A^\circ \subseteq A$
2) if $V \subseteq A$ is open then $V \subseteq A^\circ$
3) A is open $\iff A = A^\circ$

Pf 1) From definition
2) From definition
3) if $A = A^\circ$, from 1) its open
if A is open, from 2 $A \subseteq A^\circ$ and from 1 $A^\circ \subseteq A \implies A^\circ = A$ □

Def Let $A \subseteq X$, we define the closure of A ($\bar{A}$ or $cl(A)$) to be
$\bar{A} = \bigcap_{\substack{A \subseteq F \\ F \text{ closed}}} F$

Prop 1) $\bar{A}$ closed, $A \subseteq \bar{A}$
2) F closed and $A \subseteq F$ then $\bar{A} \subseteq F$
3) A is closed $\iff A = \bar{A}$

Pf trivial □

Prop the closure of $B_r(v)$ is the closed ball $\overline{B_r(v)}$

Pf Let F be the closure of $B_r(v)$
Since $\overline{B_r(v)} \supseteq B_r(v)$ and is closed, $F \subseteq \overline{B_r(v)}$
If $F \subsetneq \overline{B_r(v)}$, $\exists w \in \overline{B_r(v)}$ s.t. $w \notin F \implies v \notin B_r(v)$ so $d(v, w) = r$
Since F^c is open, $\exists \delta > 0$ s.t. $B_\delta(w) \subseteq F^c$.
Choose $t \in \mathbb{R}$ s.t. $0 < t < \min\{\delta, r\}$
Let $u = w + \frac{t}{r}(v - w)$
$\|u - w\| = t < \delta \implies u \in B_\delta(w) \subseteq F^c$
We also have $v - u = r(v - w) - \frac{t}{r}(v - w) = \frac{r - t}{r}(v - w)$
$\|v - u\| = r - t < r$, thus $u \in B_r(v) \subseteq F$ #

Def Let $A \subseteq X$, the boundary of A ∂A or $bd(A)$ is defined to be
$\partial A = \{v \in X : B_\varepsilon(v) \cap A \neq \emptyset, B_\varepsilon(v) \cap A^c \neq \emptyset, \forall \varepsilon > 0\}$

Lemma $A \subseteq X$:
1) $v \notin A^\circ \iff B_\varepsilon(v) \cap A^c \neq \emptyset$ $\forall \varepsilon > 0$
2) $v \in \bar{A} \iff B_\varepsilon(v) \cap A \neq \emptyset$ $\forall \varepsilon > 0$

Pf (1) Prove the equivalent statement $v \in A^\circ \iff \exists \varepsilon_0 > 0$ s.t. $B_{\varepsilon_0}(v) \cap A = \emptyset$
($\implies$) Trivial
($\impliedby$) Similar to $\implies$
(2) Prove the equivalent statement $v \notin \bar{A} \iff \exists \varepsilon_0 > 0$ s.t. $B_{\varepsilon_0}(v) \cap A = \emptyset$
Similar to (1) □

Corr $\partial A = \bar{A} \setminus A^\circ = \bar{A} \cap (A^\circ)^c$
(eg. $\partial B_r(v) = \{x \in X : d(x, v) = r\}$)

$\partial A = \bar{A} \cap (A^\circ)^c \implies \partial A$ is closed

Def Let $v \in X$, a neighbourhood of v is an open open set U s.t. $v \in U$

Def Let $A \subseteq X$, $A \neq \emptyset$
1) $p \in A$ is isolated iff $\exists$ a n'h'd U of p s.t. $U \cap A = \{p\}$
If $p \in A$ is isolated, $p \in \partial A$.

2) $p \in X$ is a limit point of A iff for any n'h'd U of p
$U \cap (A \setminus \{p\}) \neq \emptyset$

Remark If $v \in A^\circ$, v is a limit point and not isolated
If $v \in ext\,A$, then v is not a limit point (and not isolated)
If $p \in \partial A$, it is either an isolated point or a limit point

Def X NVS, let $(v_k)_{k \in \mathbb{N}}$ be a sequence in X.
(v_k) converges to v in X iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $k \geq N$, then $\|v_k - v\| < \varepsilon$.

Lemma Limits are unique

Prop Suppose $\lim v_k = v$. Then if U is any n'h'd of, $\exists N \in \mathbb{N}$ s.t. $k \geq N \implies v_k \in U$

Prop Let $A \subseteq X$. $v \in \bar{A} \iff \exists (v_k) \subseteq A$ s.t. $\lim v_k = v$

Def Let $(v_k) \subseteq X$. (v_k) is bounded iff $\exists M > 0$ s.t. $\|v_k\| < M$ $\forall k$.

Prop If (v_k) converges then v_k is bounded

Def (v_k) is Cauchy iff $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $\forall k, l \geq N$, $\|v_k - v_l\| < \varepsilon$

Lemma (v_k) converges $\implies (v_k)$ is Cauchy

Def X is called complete iff (Cauchy $\implies$ Converges)

Prop $v_k \to v \implies$ all subsequences of v_k converge to v

Thm (Bolzano-Weierstrass) Let (v_k) be bounded. Then it has a convergent subsequence.

Def Let X, Y NVS and $A \subseteq X$, $f: A \to Y$.
Let p be a limit point of A. We say
$\lim_{x \to p} f(x) = q \in Y$ iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t.
$v \in A$, $0 < \|v - p\| < \delta \implies \|f(v) - q\| < \varepsilon$

Lemma If $\lim_{v \to p} f(v)$ exists it is unique

Prop (SCoLoM) $f: A \subseteq X \to Y$, p limit point A. TFAE
(1) $\lim_{v \to p} f(v) = q$
(2) for any sequence $(v_k) \subseteq A \setminus \{p\}$, the sequence $(f(v_k)) \to q$

Def $f: A \subseteq X \to Y$, let $p \in A$. f is continuous at p
iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall v \in A$, $\|v - p\| < \delta \implies \|f(v) - f(p)\| < \varepsilon$.
Equivalently, $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $f(B_\delta(p) \cap A) \subseteq B_\varepsilon(f(p))$

Remark If p is an isolated point, then f is continuous at p.

Corr Let p be a limit point of A. f is cts at $p \iff \lim_{v \to p} f(v)$ exists

Def We say f is continuous of $A \iff f$ is cts at p, $\forall p \in A$

Prop Let $f: A \subseteq X \to Y$, $p \in A$
(1) f is cts at $p \iff$ for every n'h'd V of $f(p)$, $\exists$ a n'h'd U of p s.t. $A \cap U \subseteq f^{-1}(V) \iff f(A \cap U) \subseteq V$
(2) f is cts on $A \iff$ for every open subset $V \subseteq Y$, $f^{-1}(V) = A \cap U$ for some open set $U \subseteq X$

Def Let $f: A \subseteq X \to Y$. f is uniformly continuous on A iff $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\|v - w\| < \delta \implies \|f(v) - f(w)\| < \varepsilon$

Def Let $E \subseteq X$. E is disconnected iff $\exists$ open subsets U, V of X s.t.
$E \cap U \neq \emptyset$, $E \cap V \neq \emptyset$, $E \cap U \cap V = \emptyset$, $E \subseteq U \cup V$
U, V is called a seperation of E
E is connected iff $\nexists$ a seperation of E.

Recall $I \subseteq \mathbb{R}$ is an interval iff $x, y \in I$ with $x < y \implies [x, y] \subseteq I$

Thm Let $E \subseteq \mathbb{R}$. E connected $\iff E$ interval

Def Let $E \subseteq X$, E is convex $\iff \forall u, v \in E, \forall t \in [0, 1], (1-t)u + tv \in E$

Prop Convex $\implies$ Connected

Prop Let $f: E \subseteq X \to Y$ be cts on E
E connected $\implies f(E)$ connected

Cor (IVT) Let $f: E \subseteq X \to \mathbb{R}$ where f cts on E and E is connected
Let $u, v \in E$ with $f(u) < f(v)$. $\forall t \in [f(u), f(v)]$, $\exists w \in E$ s.t. $f(w) = t$

Def Let $E \subseteq X$. A open cover of E is $\{U_\alpha\}_{\alpha \in A}$ of open sets of X s.t.
$E \subseteq \bigcup_{\alpha \in A} U_\alpha$
A subcover of an open cover is a subcollection $\{U_\alpha\}_{\alpha \in B}$ which still
$E \subseteq \bigcup_{\alpha \in B} U_\alpha$

Def Let $E \subseteq X$. We say E is compact iff every open cover admits a finite subcover

Prop Any closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact.

Def Let $E \subseteq X$. E is bounded iff $\exists M > 0$ s.t. $E \subseteq B_M(0)$

Prop Compact $\implies$ Bounded

Cor Unbounded $\implies$ non Compact

Cor X is not compact

Prop Compact $\implies$ Closed

Def We say X has Heine-Borell property iff closed + bounded $\implies$ Compact

Def Let $E \subseteq X$ compact, let $F \subseteq X$ closed, $F \subseteq E \implies F$ compact

Prop Let $f: X \subseteq X \to Y$ cts on K
K compact $\implies f(K)$ compact in Y

Corr X has Hiene Borell property $\iff \overline{B_1(0)}$ is compact

Pf ($\implies$) Sps X-HB. $\overline{B_1(0)}$ is closed and bounded, thus $\overline{B_1(0)}$ compact.
($\impliedby$) Sps $\overline{B_1(0)}$ Compact. Let $M > 0$, then $f: X \to X$ $x \mapsto Mx$ is cts on X
Hence $f(\overline{B_1(0)}) = \overline{B_M(0)}$ is cpt. Let E be closed and bounded $\exists M$ s.t. $E \subseteq \overline{B_M(0)}$
By prop, E cpt. □

Lemma Let $E \subseteq X$ be compact and $U \subseteq X$ open. $\exists \delta > 0$ s.t. if $p \in E$ then $B_\delta(p) \subseteq U$

Theorem Let $R > 0$, $\overline{B_R(0)} \subseteq \mathbb{R}^n$ is cpt.

Corr $\mathbb{R}^n$ has the HB property.